Trainings

Preparatory Steps

⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lb]. To prevent serious personal injury, be sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

⚠ CAUTION ⚠

The two cam idler gears, the two camshaft gears, and the crankshaft gear have index marks that are aligned at assembly. Improper alignment will cause extensive engine damage.

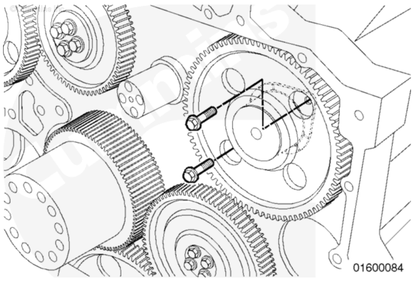
Note : Proper support of the engine is required due to the front engine mount being removed. Do **not** work on, around, or under an improperly supported or suspended engine.

- Remove the gear cover and all related components. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)
- Remove the cam follower assemblies from the necessary bank. Refer to Procedure 004-001 in Section 4. (/qs3/pubsys2/xml/en/procedures/56/56-004-001-tr.html)
- Remove one injector per bank to allow timing to be checked. Refer to Procedure 006-026 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-026-tr.html)
- Bar the engine over to align the index marks.
- Remove the left-bank camshaft compound-idler gear. Refer to Procedure 001-037 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-037-tr.html)
- Remove the accessory drive idler gear.

Note : The right bank camshaft can be removed without removing the right-bank idler gear.

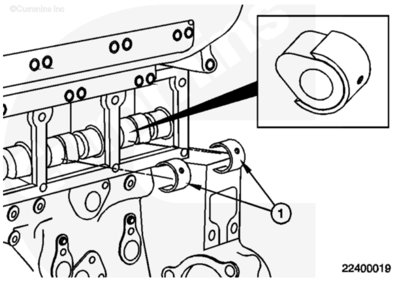
Remove

Remove the two capscrews from the thrust bearing plate through the openings in the camshaft gear.



⚠ CAUTION ⚠

Pilots must be used to prevent damage to the camshaft and the bushings.



Use camshaft tool kit, Part Number 3163195. Install the c-clips **only** on the inner base circle of the valve lobes as defined and center the clips on the camshaft. Install all clips before removing or installing the camshaft.

The c-clip locations can be seen in the chart below.

Camshaft Guide Installation Chart			
QSK 45 Camshaft Lobes		**QSK60 Camshaft Lobes**	
Left Bank	Right Bank	Left Bank	Right Bank
Exhaust - 2, 4, 5, and 6	Exhaust - 2, 3, and 5	Exhaust - 4, 6, and 8	Exhaust - 2, 4, and 7
Intake - 2 and 5	Intake - 2, 3, and 5	Intake - 2, 4, and 8	Intake - 2, 3, and 4

- Move the camshaft forward enough to enable installation of the camshaft support on the rear of the camshaft.

⚠ WARNING ⚠

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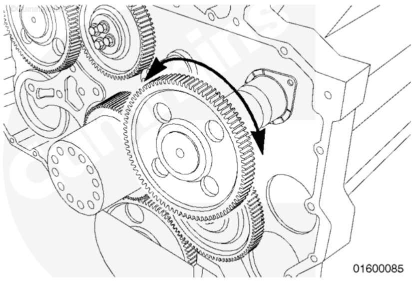


⚠ CAUTION ⚠

To reduce the possibility of damage to the camshaft or bushings during disassembly, turn the camshaft gently clockwise and counterclockwise slightly while pulling it out.

⚠ CAUTION ⚠

Rotating the camshaft too far in any direction will cause the camshaft tools to lock up the camshaft.

Note : One of the pilots **must always** be on the bottom part of the camshaft bushing.

Slightly rotate the camshaft back and forth during removal, if necessary.

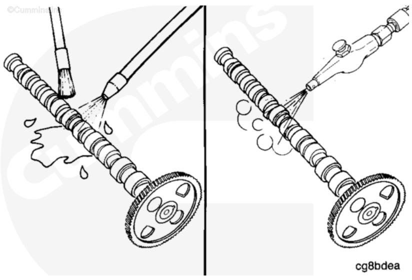
Once the camshaft extends approximately half the length of the camshaft, use a lift strap to support the camshaft and complete the removal.

Remove the camshaft from the cylinder block.

Clean and Inspect for Reuse

⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

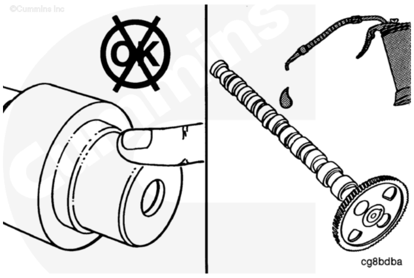
Use solvent to clean the camshaft assembly.

Dry the camshaft with compressed air.

⚠ CAUTION ⚠

To prevent rust from forming, do not touch the camshaft machined surfaces after the camshaft has been cleaned.

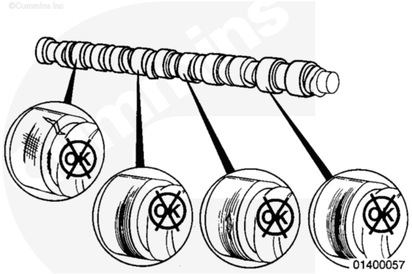
Before handling, lubricate the camshaft with clean 15W-40 oil.



Check the camshaft lobes and journals for cracks, scratches, or other damage. If the lobes or journals are damaged, the camshaft **must** be replaced.

Cummins Inc. does **not** recommend repairing camshafts by grinding the valve or injector lobes.

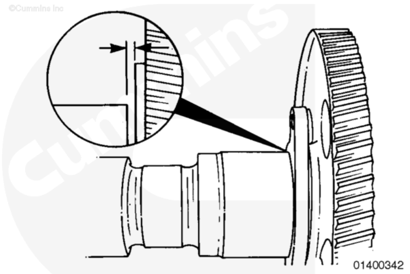
Marks on the lobes or journals that can **not** be felt with a fingernail are acceptable.



Using a feeler gauge, measure the camshaft's thrust end clearance.

Camshaft Thrust End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013

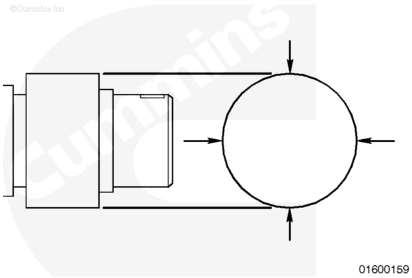
If the end clearance is **not** within specifications, the gear **must** be removed. Refer to Procedure 001-013 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-013-tr.html)



Measure the diameter of the seven camshaft bushing journals.

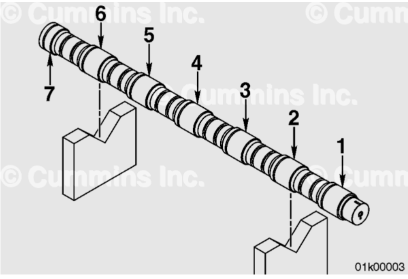
Camshaft Bushing Journal Diameter		
mm		in
104.987	MIN	4.133
105.013	MAX	4.134

If the bushing journal diameter is **not** within specifications, the camshaft **must** be replaced.



⚠ WARNING ⚠

This component or assembly weighs greater than 23 kg [50 lbs]. To prevent serious personal injurty, make sure to have assistance or use appropriate lifting equipment to lift this component or assembly.

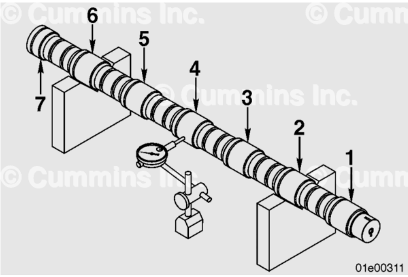


Place two vee blocks on a flat surface.

Support the camshaft on the vee blocks at the two journals indicated below.

- QSK45 camshafts: journals 2 and 6
- QSK60 camshafts: journals 2 and 7.

Position a dial indicator so the stem touches the center line of the journal. Rotate the camshaft and measure the total indicator run-out for each journal.



The dial indicator **must** be positioned at the centerline of any journal that is measured.

Max Journal Run-out		
mm		in
0.08	MAX	0.003

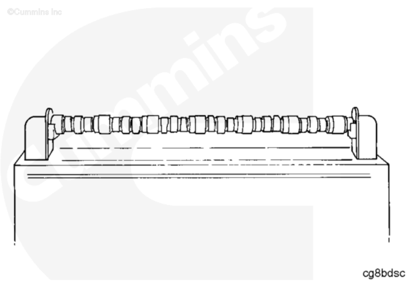
If the camshaft is **not** within specifications, it **must** be replaced.

Note : The max journal run-out applies to each journal.

Note : The two supported journals do not need to be measured.

Magnetic Crack Inspect

The gear and the thrust plate **must** be removed before performing this task. Refer to Procedure 001-013 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-013-tr.html)



Use a magnetic particle testing machine.

Using the head shot method:

- Adjust the machine to 2,000 amps direct current of rectified VAC.
- Use the continuous method. Do **not** wet more than 1/3 of the camshaft at a time.

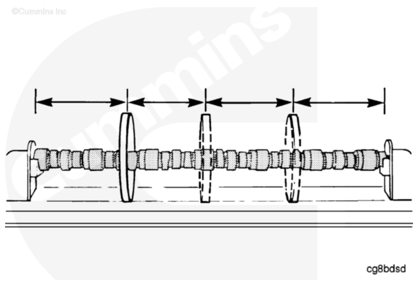
- Check the camshaft for cracks.

When the coil shot method is used, use a coil that is a minimum of 305 mm [12 in] diameter.

Ampere turn is an electrical current of one ampere flowing through the coil, multiplied by the number of turns in the coil.

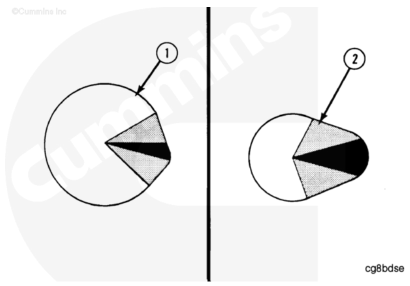
Use a coil with a minimum of 3,600 amps direct current or rectified VAC and a maximum of 4000 amps direct current or rectified VAC.

Check the camshaft for cracks.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.



An open indication is visible after the wetting operation has been completed. An indication below the surface is **not** visible after the wetting operation has been completed. An indication below the surface is visible with the use of ultraviolet light.

Do **not** use the camshaft if the bearing journals have:

Bearing Journal Limits of Acceptance

- More than four open indications in an axial direction on any one bearing journals.
- More than one half of the open indications extend more than one half of the distance across the bearing journals.
- Open indications in a circumferential direction.

The camshaft can be used if there are indications below the surface of the bearing journals.

Do **not** use the camshaft if the lobes have:

Camshaft Lobe Limits of Acceptance - Open Indications

- Open Indications in a circumferential direction.
- Open Indications in the black or shaded areas (2).
- Open indications that are longer than 6 mm [0.236 in].
- Open indications that are closer than 5 mm [0.197 in] from the edge.
- More than two indications on one of the lobes.

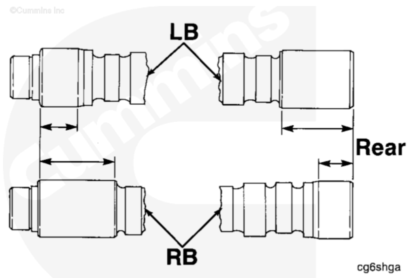
Camshaft Lobe Limits of Acceptance - Indications Below the Surface

- More than two indications in the shaded area (2).
- Below the surface indications in shaded area (2) that are longer than 16 mm [0.630 in].
- More than three indications in the white area (1).
- Below the surface indication in the black area (2) on the injector area.
- More than two indications in the black area on the valve lobe (2).
- Below the surface indications in the black area (2) of the valve lobe that are longer than 3 mm [0.118 in].

Demagnetize the camshaft.

Clean the camshaft with solvent.

Install

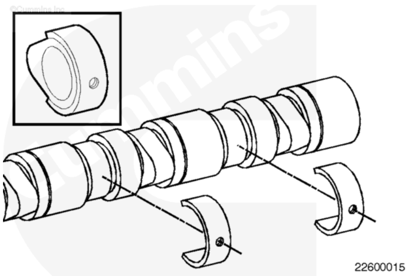


⚠ CAUTION ⚠

The right and left camshafts are different. They must be installed in the correct bank. If they are not installed correctly, the engine will be seriously damaged.

Note : The camshafts for the QSK45 and QSK60 are different. The installation procedure, however, is the same.

Make sure the correct camshaft is installed. Refer to the CPL for the engine being serviced. If the wrong camshaft is installed, the engine will **not** perform to specifications.

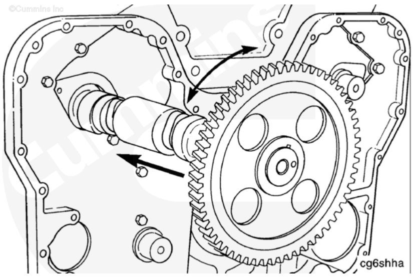


Use camshaft tool kit, Part Number 3163195. Install the camshaft installation guides **only** on the inner base circle of the valve lobes as defined and center the guides on the camshaft. Install all of the guides before removing or installing the camshaft.

Install the camshaft installation guides, Part Number 3163195, on the valve lobes of the camshaft.

Install the camshaft pilot onto the rear of the camshaft.

Use clean engine oil to lubricate the camshaft and camshaft bushings.



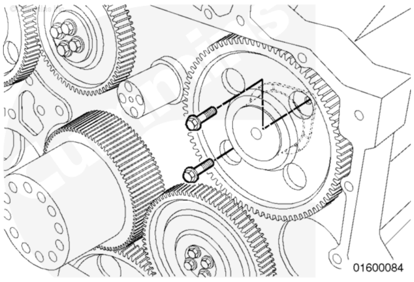
⚠ WARNING ⚠

This component weighs 23 kg [50 lb] or more. To reduce the possibility of personal injury, use a hoist or get assistance to lift this component.

⚠ CAUTION ⚠

Do not damage the camshaft bushings when installing the camshaft. The bushing will fail.

Install the camshaft by turning it slightly **clockwise** and **counterclockwise** as it is pushed into the block. Keep the guides turned downward to support the camshaft.

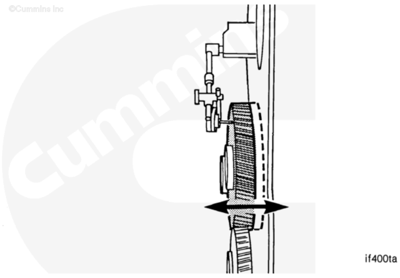


Align the thrust bearing, and install the two capscrews.

The narrow side of the thrust bearing plate is installed towards the outside of the engine block.

Tighten the capscrews.

Torque Value: 45 n•m [33 ft-lb]

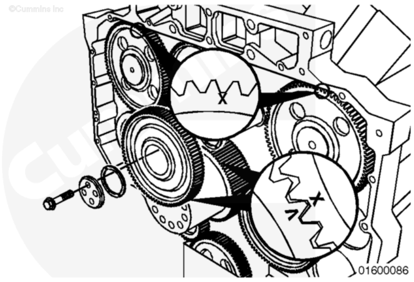


Use a dial indicator, Part Number 3376050, and magnetic base, Part Number 3377399, to measure the camshaft's end clearance.

Camshaft End Clearance		
mm		in
0.15	MIN	0.006
0.33	MAX	0.013

If the clearance is **not** within specifications, check for foreign material or a piece of gasket between the thrust plate and the block.

Finishing Steps



Make sure the index marks on the camshafts are in the 12 o'clock position and crankshaft gear and the idler gears are in alignment with the timing marks.

Install the left-bank camshaft compound idler gear.

- Measure the gear train backlash. Refer to Procedure 001-055 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-055.html)
- Install the cam follower assembly on the cylinder that previously had the injector removed. Refer to Procedure 004-001 in Section 4. (/qs3/pubsys2/xml/en/procedures/56/56-004-001-tr.html)
- Measure the injection timing. Refer to Procedure 006-025 in Section 6. (/qs3/pubsys2/xml/en/procedures/56/56-006-025-tr.html)
- Install the remaining cam follower assemblies and related components. Refer to Procedure 004-001 in Section 4. (/qs3/pubsys2/xml/en/procedures/56/56-004-001-tr.html)
- Install the gear cover and related components. Refer to Procedure 001-031 in Section 1. (/qs3/pubsys2/xml/en/procedures/56/56-001-031-tr.html)